

# ETIENNE NZABARUSHIMANA, PhD, MPH

Senior Computational Biologist | Bioinformatics Scientist | Translational Multi-Omics | Biomarker Discovery | Machine Learning

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## PROFESSIONAL SUMMARY

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Computational biologist and bioinformatics scientist with 10+ years of experience applying statistical, machine-learning, and bioinformatics methods to large-scale genomic, multi-omic, and human clinical-cohort data. PhD trained in Bioinformatics and Computational Biology, with graduate training in biostatistics and an MPH, and postdoctoral research across Massachusetts General Hospital, Harvard Medical School, and Harvard T.H. Chan School of Public Health. Expertise in biomarker and molecular-signature discovery, patient and cohort stratification, high-dimensional data analysis, multi-omics integration, disease-mechanism analysis, and reproducible computational method development. Experienced in translating biological and clinical questions into rigorous analytical strategies, scalable workflows, interpretable results, and cross-functional scientific deliverables.

## CORE CAPABILITIES

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**Translational Computational Biology:** Biomarker discovery; molecular-signature discovery; clinical-cohort analysis; patient stratification; disease-mechanism analysis; translational research; experimental and NGS study design; molecular and clinical data integration.

**Statistical Modeling, Machine Learning & AI:** Classification; logistic and penalized regression; feature selection; predictive modeling; cross-validation; model evaluation; high-dimensional data analysis; confounder-aware analysis; interpretable biological modeling.

**Programming, Data Science & Workflows:** Python; R; Bash; Linux/Unix; Git; scientific Python; Bioconductor; Biopython; statistical computing; data wrangling and visualization; bioinformatics pipeline development; workflow automation; reproducible notebooks; high-performance computing; parallel and batch analysis; quality control and documentation.

**Multi-Omics & Disease Biology:** Genomics; transcriptomics; metagenomics; metatranscriptomics; viral profiling; metabolomics; functional and pathway profiling; epigenetics; inflammatory and metabolic disease; cancer-risk biology; host-microbe biology; mechanistic interpretation.

## PROFESSIONAL EXPERIENCE

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### Postdoctoral Research Fellow, Computational Biology & Bioinformatics

Massachusetts General Hospital, Harvard Medical School, & Harvard T.H. Chan School of Public Health | Boston, MA | August 2022 - July 2026

- Led computational analyses of large-scale human-cohort multi-omics and clinical datasets, translating biological and clinical questions into reproducible analytical strategies, statistical models, visualizations, and publication-quality scientific outputs.
- Integrated metagenomic, metatranscriptomic, viral, metabolomic, genomic, and clinical data using statistical modeling, machine learning, feature selection, and pathway-level analysis to identify molecular signatures, biomarkers, patient subgroups, and disease mechanisms.
- Developed reusable Python, R, Bash, and Linux-based workflows for data processing, quality control, integration, statistical analysis, visualization, and scalable HPC execution using Git, workflow automation, documented code, and reproducible analysis practices.
- Co-developed WAAFL, a computational method for detecting lateral gene transfer events in metagenomes; applied the method to Human Microbiome Project data and co-led the resulting *Nature Microbiology* publication.
- Contributed computational analyses to multi-omic human-cohort studies published in *Nature Metabolism*, *Nature Communications*, and *Gastroenterology*, working with clinicians, epidemiologists, molecular biologists, bioinformaticians, and computational scientists.

### Research Assistant, Bioinformatics

Indiana University Bloomington | Bloomington, IN | August 2015 - July 2022

- Developed algorithms and analytical workflows for large-scale sequencing, human genomics, microbial genomics, genome structure, molecular-signature discovery, and high-dimensional biological data analysis.
- Analyzed 10,000+ human metagenomic samples spanning *Clostridioides difficile* infection, IBD, IBS, celiac disease, constipation, and diarrhea cohorts to identify taxonomic, functional, and molecular features associated with disease states.
- Developed and validated machine-learning models for disease classification and cohort stratification using high-dimensional sequencing-derived features, including a published CDI classifier with average cross-validated AUC  $\geq 0.91$  and a compact disease-associated microbial signature.
- Developed STRScan for targeted short-tandem-repeat profiling from whole-genome sequencing data; contributed to clonal-reconstruction algorithm development and designed methods for CRISPR-array detection in metagenomic data.

### Research Assistant, Molecular & Epigenetic Biology

University of Arkansas for Medical Sciences | Little Rock, AR | January 2012 - July 2014

- Led molecular and epigenetic studies of ionizing-radiation effects relevant to cancer risk, lung biology, fibrosis, inflammatory signaling, transcriptional regulation, and disease mechanisms using cell-based and mouse-model experiments.

- Generated and interpreted gene/protein expression, flow cytometry, microscopy, HPLC, mass spectrometry, and related assay data; identified epigenetic biomarkers associated with radiation-induced transcriptional repression, contributing to 2 first-author and 3 co-authored publications.

## EDUCATION

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### PhD in Informatics - Bioinformatics & Computational Biology, May 2022

Indiana University Bloomington, Bloomington, IN

### Master of Public Health in Biostatistics, Honors, May 2014

University of Arkansas for Medical Sciences, Little Rock, AR

### Master of Science in Informatics, May 2021

Indiana University Bloomington, Bloomington, IN

### Bachelor of Arts in Chemistry, Magna Cum Laude, May 2013

Hendrix College, Conway, AR

## SELECTED PUBLICATIONS

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- Hsu, T.Y., **Nzabarushimana, E.**, Wong, D., Luo, C., Beiko, R.G., Langille, M., Huttenhower, C., Nguyen, L.H., Franzosa, E.A. Profiling lateral gene transfer events in the human microbiome using WAAFL. *Nature Microbiology*. 2025;10(1):94-111. Co-lead author.
- Kim, H., Nelson, P., **Nzabarushimana, E.**, et al. Multi-omic analysis reveals transkingdom gut dysbiosis in metabolic dysfunction-associated steatotic liver disease. *Nature Metabolism*. 2025;7(7):1476-1492.
- Chen, A.S., Kim, H., **Nzabarushimana, E.**, et al. Association of distinct microbial and metabolic signatures with microscopic colitis. *Nature Communications*. 2025;16(1):4644.
- **Nzabarushimana, E.** and Tang, H. Functional profile of host microbiome indicates *Clostridioides difficile* infection. *Gut Microbes*. 2022;14(1):2135963.
- **Nzabarushimana, E.**, Prior, S., Miousse, I.R., et al. Combined exposure to protons and <sup>56</sup>Fe leads to overexpression of IL13 and reactivation of repetitive elements in mouse lung. *Life Sciences in Space Research*. 2015;7:1-8.

## SCIENTIFIC LEADERSHIP & SERVICE

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- Instructor, Microbial Community Multi-omics using the bioBakery, Harvard University, 2025; guest lecturer, The Microbiota-Gut-Brain Axis, Harvard University, 2024.
- Instructor of Record and Associate Instructor, Data Science in Practice, Indiana University, 2019-2021; mentored junior researchers in computational analysis, interpretation, and scientific communication.
- Ad hoc reviewer for *BMC Bioinformatics*, *Microbiome*, *Gut Microbes*, *Bioinformatics*, *Cell Systems*, *Frontiers in Molecular Biosciences*, and *NAR Genomics and Bioinformatics*.

## ADDITIONAL STRENGTHS

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Cross-functional collaboration with clinical, epidemiology, molecular biology, and computational teams; scientific writing and manuscript development; presentation of complex analyses to technical and non-technical audiences; HIPAA and patient-data privacy awareness; reproducible and version-controlled analysis practices.